

MAHNOOR ZAFFAR

AI ENGINEER | PYTHON SPECIALIST

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Strategic AI Engineer and Python Specialist with 3+ years of cross-functional experience in FinTech and HealthTech. Expert in bridging the gap between complex machine learning models and production-ready Full-Stack applications. Specialised in **LLM Orchestration (RAG), MLOps, and Scalable Backend Architectures**. Proven track record of delivering high-impact AI solutions that reduce latency by 40% and drive measurable ROI through automated decision-making and predictive analytics.

AREA OF EXPERTISE

-Backend Engineering & Distributed Systems | AI/ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain | DevOps, Cloud & Deployment Automation | Applied AI & Intelligent Product Development | **Data Engineering** | **MLOps** & Cloud | Generative AI: RAG (Retrieval-Augmented Generation), Prompt Engineering, Fine-tuning (LoRA/QLoRA), OpenAI/Anthropic API integration | Computer Vision

KEY ACHIEVEMENTS

AUGUST 2021

Identified by **Affirm Holdings, Inc. [an American financial technology]** for a software engineering role in **2021 based on strong technical performance, problem-solving ability, and clear growth trajectory.**

Although I was unable to proceed due to bereavement leave, the invitation affirmed my potential as a high-impact engineer with the capacity to perform at top-tier, product-driven companies.

PROFESSIONAL EXPERIENCE

MAY 2019 - PRESENT

FinTrack AI: Predictive Risk Scoring & GTM Intelligence

Architected a full-stack Python application for the Pakistani freelance market to predict creditworthiness using alternative data.

- Developed a Random Forest and **XGBoost ensemble model** to analyze non-traditional financial indicators, achieving a 92% prediction accuracy.
- Built the backend using **FastAPI** and **PostgreSQL**, handling 50k+ concurrent data points with sub-100ms latency.

MediSync RAG: HIPAA-Compliant Medical Knowledge Base

Built a Retrieval-Augmented Generation (RAG) system for HealthTech providers to query internal medical research.

- Implemented a **vector search engine using Pinecone and LangChain**, integrating a fine-tuned Llama-3 model for medical-grade accuracy.
- Engineered a React/**Django interface** that streams AI responses via WebSockets, reducing clinician research time by 60%.

EDUCATION

Back-end Engineer | Self-Taught Software Development | Codecademy Professional Cert.

Engineered 40+ full-stack projects through intensive self-study. Mastered Algorithms, Backend Systems, and Scalable Programming via 40+ specialized certifications. Expert at rapid upskilling and independent technical execution in industry-standard technologies.

BSc (Hons) Biotechnology | Fatima Jinnah Women University

October 2018 - October 2022

ADDITIONAL INFORMATION

Certifications: **Professional Certificate in AI Engineering (2026)** | Coursera

Associate Python For Developers (Complete Roadmap) | DataCamp

Back-End Developer | Codecademy